1. Outline.

2. A coupled partial melting Stokes-Darcy system. Strong form

$$
\begin{aligned}
\tilde{\mathbf{v}}_r + \frac{k_0\phi_f^{1+\Theta}}{\mu_f}\nabla(\phi_f^{-1/2}\tilde{q}_f) &= 0\\
\mu_s\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2}q) &= 0\\
\nabla q - \nabla\cdot\hat{\sigma}(\mathbf{v}_s) &= -(1-\phi_f)\rho_r\mathbf{g}\\
\mu_s\nabla\cdot\mathbf{v}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2}q) &= 0
\end{aligned}
\tag{2.1}
$$

List SI units
- q = pascal.
- $\mathbf{v}$ = m/s

Non dimensionalization choice
- $\check{\mathbf{v}} = \frac{\tilde{\mathbf{v}}_r}{k_0\rho_r g_y/\mu_f}$
- $\check{q} = \frac{q}{\rho_r g_y l_0}$
- $\check{x} = \frac{x}{l_0}$
- $l_0 = (\frac{k_0\mu_s}{\mu_f})^{1/2}$

The resulting non dimensionalized form is therefore

$$
\begin{aligned}
\check{\mathbf{v}}_r + \phi_f^{1+\Theta}\nabla(\phi_f^{-1/2}\check{q}_f) &= 0\\
\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0\\
\nabla\check{q} - \nabla\cdot\hat{\sigma}(\check{\mathbf{v}}_s) &= -(1-\phi_f)\rho_r\hat{\mathbf{g}}\\
\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0
\end{aligned}
\tag{2.2}
$$

scaled nondimensionalized weak form let $\check{\sigma} = \mathcal{D}\mathbf{v} - \frac{1}{3}\nabla\cdot\mathbf{v}\mathbf{I}$

$$
\begin{aligned}
(2.3) \qquad & (2\phi_s \check{\sigma}(\mathbf{v}_{s,h}), \nabla \Psi_s) - (\check{q}_h, \nabla \cdot \Psi_s) = -(\phi_s \check{\mathbf{f}}, \Psi_s) \\
& (\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s} \check{q}_h, \omega\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \check{q}_{f,h}, \omega\right) = 0 \\
& (\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \check{\mathbf{v}}_r)) = 0 \\
& (\hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \check{\mathbf{v}}_r), \omega_f) + \left(\frac{1}{\phi_s} \check{q}_{f,h}, \omega_f\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \check{q}_h, \omega_f\right) = 0
\end{aligned}
$$

2.1. Variational form and discretization.

2.1.1. Stokes and variational formulation.

$$
\begin{aligned}
-\Delta \mathbf{u} + \nabla p &= \mathbf{f} \\
\nabla \cdot \mathbf{u} &= 0
\end{aligned} \tag{2.4}
$$

weak form

$$
\begin{aligned}
(\nabla \mathbf{u}, \nabla \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{f}, \mathbf{v})_\Omega + \langle \tau^T \nabla \mathbf{u} \nu, \mathbf{v} \cdot \tau \rangle_{\Gamma_i} \\
&\quad + \langle \nu^T \nabla \mathbf{u} \nu - p_0, \mathbf{v} \cdot \nu \rangle_{\Gamma_i} - \langle \nabla \mathbf{u}_0, \nabla \mathbf{v} \rangle_{\Gamma_u} \\
(\nabla \cdot \mathbf{u}, \omega) &= 0
\end{aligned} \tag{2.5}
$$

let stress $\mathbf{s} \cdot \tau = \tau^T \nabla \mathbf{u} \nu$ and $\mathbf{s} \cdot \nu = \nu^T \nabla \mathbf{u} \nu - p_0$, we have traction boundary condition term defined on Γ_i which is $< \mathbf{s}, \mathbf{v} >_{\Gamma_i}$. Traditionally a test space $\{\mathbf{v} \in H^1(\Omega) : \mathbf{v} = 0 \text{ on } \Gamma_u\}$ is employed here.

$$
\begin{aligned}
(\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} &= (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + \langle \mathbf{s}, \mathbf{v}_h \rangle_{\Gamma_{i,h}} - \langle \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h \rangle_{\Gamma_{u,h}} \\
(\nabla \cdot \mathbf{u}, \omega) &= 0
\end{aligned} \tag{2.6}
$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

2.2. Darcy.

$$
\begin{aligned}
\mathbf{u} + \nabla p &= \mathbf{g} \\
\nabla \cdot \mathbf{u} &= 0
\end{aligned} \tag{2.7}
$$

weak form

$$
\begin{aligned}
(\mathbf{u}, \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{g}, \mathbf{v})_\Omega - \langle p_0, v_n \rangle_{\Gamma_i} - \langle \mathbf{u}_0, \mathbf{v} \rangle_{\Gamma_u} \\
(\nabla \cdot \mathbf{u}, \omega) &= 0
\end{aligned} \tag{2.8}
$$

Test space $\{\mathbf{v} \in H(\text{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

2.2.1. A degenerate Darcy test problem.

$$
\begin{aligned}
\mathbf{v} + \phi \nabla (\hat{\phi}^{-1/2} p) &= f \\
\hat{\phi}^{-1/2} \nabla \cdot (\phi \mathbf{v}) &= g
\end{aligned} \tag{2.9}
$$

We manufacture a solution with degenerate porosity in the target physical domain $x, y \in [-1, 1]$.

$$
\phi = \begin{cases} \cos^m(n(x^2 + y^2)) & x^2 + y^2 < \pi/2n \\ 0 & \text{otherwise} \end{cases} \tag{2.10}
$$

where m is an even number, and $\pi/2n < 1$ another choice of test case is shown as follows:

$$
\phi = \begin{cases} 0 & x \leq 0 \\ \phi_+ x^4 & x > 0 \end{cases} \tag{2.11}
$$

Define an arbitrary velocity and pressure

$$\begin{aligned}\mathbf{v} &= [\frac{x}{5}, -y]^T \\ p &= \hat{\phi}^{1/2}\end{aligned} \tag{2.12}$$

The source term is calculated as follows:

$$f = \mathbf{v} \tag{2.13}$$

and the divergence term is:

$$\begin{aligned}g &= \phi^{-1/2}\phi_{+}\nabla\cdot[\frac{x^5}{5}, -x^4 y]^T \\ &= \phi^{-1/2}\phi_{+}(x^4 - x^4) = 0\end{aligned} \tag{2.14}$$

a divergence free test case

2.2.2. Modified BR element. The Bernardi-Raugel element has been researched extensively in previous literature [2]. The error estimate is well known established by[4].

$$\|\mathbf{u}-\mathbf{u}\|_{1,\Omega}+\|p-p_h\|_{0,\Omega} \leq Ch^m(\|\mathbf{u}\|_{m+1,\Omega}+\|p\|_{m,\Omega}) \tag{2.15}$$

We state that our new function space satisfies the following hypotheses such that the error estimate holds:

- For any q in $H^m(\Omega)\cap L_0^2(\Omega)$, one has

$$\inf_{q_h\in\mathbf{M}_h}\|q-q_h\|_{0,\Omega} \leq Ch^m\|q\|_{m,\Omega}$$

- there exists a linear operator Π_h from $H^{m+1}(\Omega)^d\cap H_0^1(\Omega)^d$ into X_h such that

$$\forall \mathbf{v}\in H^{m+1}(\Omega)^d\cap H_0^1(\Omega)^d, \begin{cases} \forall q_h\in M_h, (q_h, \mathrm{div}(\mathbf{v}-\Pi_h\mathbf{v}))=0, \\ \qquad \|\mathbf{v}-\Pi_h\mathbf{v}\| \leq Ch^m\|\mathbf{v}\|_{m+1,\Omega} \end{cases}$$

- for each q_h in M_h, there exists a function $\mathbf{v}_h$ in X_h such that

$$(\mathrm{div}\mathbf{v}_h, q_h) \geq \beta\|q_h\|_{0,\Omega}\|\mathbf{v}\|_{1,\Omega}$$

  , where $\beta > 0$ is a constant independent of h.

We introduce a family of quadrilateral mesh of domain Ω, bounded by h.

3. Linear system and Uzawa type algorithm. Boundary conditions are already included in the linear system formulation. Original linear system:

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix} \tag{3.1}$$

This is a double saddle point system. Symmetrically permutate original linear system and form a enlarged saddle point system.

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix} \tag{3.2}$$

(3.3)

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}$$

In this newly formed system, the $\mathbf{A}$ matrix maintains as a positive definite matrix. we solve this typical saddle point system with uzawa iteration. Uzawa algorithm has been discussed extensively at [3], [6].

Consider the linear system of equation:

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix} \tag{3.4}$$

Algorithm 3.1 Preconditioned Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
3: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \mathbf{Q}^{-1}(\mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G})$

In the previous literature [6], $\mathbf{Q}$ is approximated with $\tau\mathbf{I}$, which works well with the stokes part. However, it converges very slow with the same approximation when applied to the Darcy part. We need a better preconditioner approximating inverse of the corresponding schur complement. We use MINRES algorithm [1] due to schur complement being symmtetric and indefinite.

Algorithm 3.2 Preconditioned Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
3: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
4: Obtain $\tilde{\mathbf{y}} = \operatorname{argmin}_{\mathbf{x}\in V} ||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$
5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

Based on the above algorithm, we proposed a new block preconditioner for our coupled linear system.

$$\mathbf{Q}^{-1} = \begin{bmatrix} \mathbf{S}_s^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_d^{-1} \end{bmatrix} \tag{3.5}$$

where the schur complement matrices are approximated with MINRES method.

3.1. Convergence and stability analysis. Uzawa iteration is in fact a fixed point iteration procedure. Usually, uzawa iteration convergences slowly.

3.1.1. Anderson acceleration. Accelerating uzawa iteration has been discussed in the previously literatures. Anderson acceleration (Anderson mixing) [9]

4. Test cases. We test the linear system a little bit different from the linear system above.

4.1. Constant porosity and balanced pressure (divergence free velocity field).

4.2. Const porosity and unbalanced pressure.

$$\check{q} = 0 \qquad \check{q}_f = 2(x+y) \qquad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \qquad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \tag{4.1}$$

$$\nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1-\phi_f}(x+y) \tag{4.2}$$

$$\mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \qquad (\nabla \cdot \check{\mathbf{v}}_s)\mathbf{I} = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x+y & 0 \\ 0 & x+y \end{bmatrix} \tag{4.3}$$

$$\begin{aligned}
\tilde{\mathbf{v}}_r + \phi_f^{1/2}\nabla\check{q}_f &= -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2}\begin{pmatrix} 2 \\ 2 \end{pmatrix} \\
\phi_f^{1/2}\nabla \cdot (\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}\check{q}_f &= \frac{-2}{1-\phi_f}(x+y) + \frac{1}{1-\phi_f}2(x+y) = 0 \\
\nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}\check{q}_f &= \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) - \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) = 0 \\
-2(1-\phi_f)\nabla \cdot \sigma(\check{\mathbf{v}}_s) &= -2(1-\phi_f)\nabla \cdot (\mathcal{D}\check{\mathbf{v}}_s - \frac{1}{3}(\nabla \cdot \check{\mathbf{v}}_s)\mathbf{I}) = -\frac{8\phi_f^{1/2}}{3}\begin{pmatrix} 1 \\ 1 \end{pmatrix}
\end{aligned} \tag{4.4}$$

5. Retrieve of physical variables. Eventually, we need to retrieve actual physicla values from nondimensionalized scaled variables. We need original velocity and pressure.

As stated above, the nondimensionalized scaled pressure potential $\check{q} = \frac{q}{\rho_r g_y l_0}$, hence $q = \rho_r g_y l_0 \check{q}$. The pressure potentials eliminated the effect of gravity, where $q_f = p_f - \rho_f g z$ and $q_s = p_s - \rho_f g z$. We can retrieve of original velocity with the similar manner $\tilde{\mathbf{v}} = \phi_f^{-1-\Theta}\mathbf{v}_r$.

6. Finite Volume with MLWENO reconstruction. A general scalar advection-diffusion equation can be represented as follows:

$$u_t + \nabla\cdot(F(u) - D(u)\nabla u) = 0,\ \mathbf{x}\in\mathbb{R}^d \tag{6.1}$$

which can be discretized with finite volume scheme.

$$\bar{u}_{E,t} + \frac{1}{|E|}\int_{\partial E}(\hat{f}(u) - \nabla u\cdot \nu_E)d\sigma(\mathbf{x}) = 0 \tag{6.2}$$

where piece-wise constant solution $\bar{u}_E = \frac{1}{|E|}\int_E u(\mathbf{x},t)d\mathbf{x}$. We choose Lax-Friedrichs scheme to represent our numerical flux scheme. We approximate integrals over each facet with quadrature rule.

$$\bar{u}_{E,t} + \sum_{j=1}^{L_E}\frac{e_j}{|E|}\sum_k \omega_{j,k}[\hat{f}(\mathscr{R}^+,\mathscr{R}^-) - \mathscr{D}(u)] = 0$$

where $\mathscr{D}$ denotes an approximation to diffusion flux along normal direction, and $\mathscr{R}$ represents a MLWENO reconstruction of values on the edge.

7. Eutectic phase behavior. Mid-ocean ridge forms at the place where temperature is generally lower. The binary eutectic phase behavior describes the chemical behavior of two immiscible crystals from a completely miscible solution.

On a typical eutectic phase behavior diagram, we can locate any phase situation using temperature and composition information. We follows phase behaviors details described in [8]. We made some reasonable adjustments for our mantle system.

There are few key elements that allow practical calculation of phase region and corresponding volumetric fractions.

- Fluid formed by eutectic melting has a constant composition of opx denoted as X_e. The eutectic composition can be related to pressure and denoted as $X_e(P)$.
- Melting temperature relates to pressure. Decompression melting is accounted for in our model by relating eutectic melting temperature with pressure change.
- Eutectic melting is of the most insterest. The temperature where eutectic melting happens has much lower temperature then actual melting temperature.

We define following dimensionless bulk variables following [8]. We denote volumetric fraction of three phases as olivine ϕ_1, opx ϕ_2 and melt ϕ_f. We are tracking opx in the mixture for simulation. Stefan number is defined as the ratio of the sensible heat to latent heat, Ste $= \frac{c_p\Delta T}{L}$. We define dimensionless temperature with eutectic temperature as $\mathcal{T} = \frac{T-T_e}{T_{\text{MAX}-T_e}}$, and dimensionless pressure as $\mathcal{P} = \frac{P-P_0}{P_0}$, where P_0 represents standard atmospheric pressure. We use melting temperature of olivine at standard atmospheric pressure as 2053 K, and 1753 K for orthopyroxene (opx). Olivine has density range of 3.2 to 4.3 g/cm^3 and orthopyroxene has similiar density ranging from 3.2 to 4 g/cm^3. We define a dimensionless composition $\chi_i = \frac{X}{X_e}$ where $i\in\{1,2,f\}$ and bulk composition defined as $\chi = \sum\phi_i\chi_i$. The eutectic melting composition is now normalized as $\chi_e = X_e/X_e = 1$.

$$\begin{aligned} C &= \phi_2\rho_2 + \phi_f\rho_f\chi_f(\mathcal{T},\mathcal{P}) \\ H &= \phi_1\rho_1 c_{p,1}\mathcal{T} + \phi_2\rho_2 c_{p,2}\mathcal{T} + \phi_f\rho_f\left(L_f + c_{p,f}\mathcal{T}\right) \end{aligned} \tag{7.1}$$

Where the melting latent heat can be affected by multiple things. We can approximate the latent heat of the mixture as

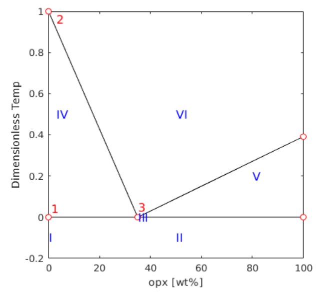
$$L_f = L_1(1-\chi_f(\mathcal{T},\mathcal{P})) + L_2\chi_f(\mathcal{T},\mathcal{P})$$

. We can nondimensionalize bulk composition and dimensionless bulk enthalpy as:

$$\begin{aligned}\mathcal{C} &= C/\rho_1 \\ \mathcal{H} &= H/(c_{p,1}\rho_1)\end{aligned} \tag{7.2}$$

We denote nondimensionalized density and specific heat as $\rho_{i,j}$ and $c_{p,i,j}$.

The eutectic phase diagram can be simplified as:



7.1. Primary phase split. In mantle, we mostly have olivine and orthopyroxene (opx) mixture in the rock. Like eutectic melting system e.g. salt and ice, the presence of one phase would largely lower the melting temperature of another phase to the eutectic melting point. We now decompose different phase regions in details We are tracking bulk concentration of orthopyroxene (opx) in our calculation. We identify following 6 regions with different phase assemblages:

I Single-phase sub-solidus : olivine
II Two-phase sub-solidus: olivine + opx
III Three-phase eutectic: olivine + opx + melt
IV Two-phase super-eutectic : olivine + melt
V Two-phase super-eutectic : opx + melt
VI Single-phase super-liquidus : melt

7.1.1. Region I. Pure olivine. The conditions to be in region I are

$$\mathcal{C} = 0 \quad \mathcal{H} \leq 1 \tag{7.3}$$

Hence, the volumetric fraction and temperature can be calculated simply as:

$$\phi_1 = 1, \quad \phi_2 = \phi_f = 0 \quad \text{and} \quad \mathcal{T} = \mathcal{H} \tag{7.4}$$

7.1.2. Region II. Two-phase sub-solidus: olivine + opx. In this phase region, temperature hasn't reached eutectic melting temperature.

$$\phi_2 = \frac{\mathcal{C}}{\rho_{2,1}} \quad \phi_1 = 1 - \phi_2 \quad \phi_f = 0 \quad \text{and} \quad \mathcal{T} = \frac{\mathcal{H}}{\phi_1 + \phi_2\rho_{2,1}c_{p,2,1}} < 0 \tag{7.5}$$

7.1.3. Region III. In this three-phase eutectic region, all three phases are present. The temperature is fixed at the eutectic point. This is the exact eutectic point where solid phase opx hasn't been exausted. In this phase region, temperature is exactly at the eutectic temperature,

$$\phi_f = \frac{\mathcal{H}}{L\rho_{f,1}} \quad \phi_2 = \frac{\mathcal{C} - \phi_f\rho_{f,1}\chi_e}{\rho_{2,1}} \quad \phi_1 = 1 - \phi_f - \phi_2 \quad \text{and} \quad \mathcal{T} = 0 \tag{7.6}$$

In this region, density of melt is fixed as eutectic density $\rho_f = \rho_e$.

7.1.4. Region IV. In this region, melting starts at the point left to the eutectic melting point. This is the melting happening in upper mantle. In this phase region, opx has exausted in solid matrix and temperature keeps rising. As more olivine melting, the density of melting changes.

$$\phi_2 = 0 \tag{7.7}$$

7.1.5. Region V. In this region, melting starts at the point right to the eutectic melting point. We are not investigate too much in this type of melting.

7.1.6. Region VI. In this region, only melting is present. Hence the volume fraction is given:

$$\phi_1 = \phi_2 = 0, \quad \phi_f = 1 \tag{7.8}$$

7.2. Further approximation and simplified phase split. To further simplify my calculation of phase diagram, I propose following approximations:

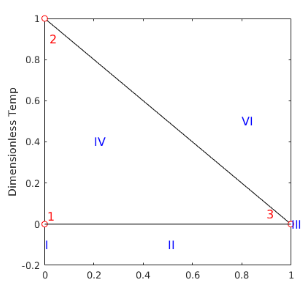
- We approximate both olivine and orthopyroxene with one uniform constant density.
- Heat capacity has a complex dependency on temperature, pressure and mixture. We assume the same specific heat capacity under circumstance of mantle melting.
- Ignore both thermal expansion and density change upon phase transition.
- Assume olivine and orthopyroxene share one same constant latent heat $L_1 = L_2 = L$.
- liquid and solid density ....

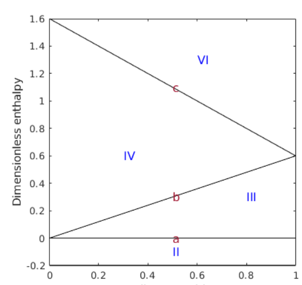
Under such assumption, we can get a much simplified concentration and bulk enthalpy:

$$\begin{aligned} \mathcal{C} &= \phi_2 + \phi_f \chi_f(\mathcal{T}, \mathcal{P}) \\ \mathcal{H} &= (\phi_1 + \phi_2 + \phi_f)\mathcal{T} + \phi_f L_f \\ &= \mathcal{T} + \phi_f L \end{aligned} \tag{7.9}$$

7.2.1. Phase diagrams. In practical, we are focusing on eutectic melting happening when mass fraction of opx is left to eutectic point. We notice that upper mantle consists of more than 80% of olivine according to[11]. Hence, we only need to restrict our calculation to the left side of eutectic point.

We can also plot a $\mathcal{H} - \chi$ diagram based on the $\mathcal{T} - \chi$ plot above.

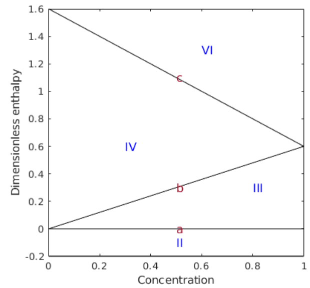




More importantly, we need $\mathcal{H} - \mathcal{C}$ phase diagram. Now we calculate the line b separate region III (eutectic melting) and region IV (super-eutectic melting). Along line b, all opx has been exhausted and eutectic melting has stopped, hence $\phi_2 = 0$. The resulting $\mathcal{C} = \phi_f \chi_f$. According to our assumption of homogenous density, at the solidus region, we have the following relationship: $\chi = \frac{\phi_2}{\phi_2 + \phi_1} = \frac{M_2}{M_1 + M_2}$. Therefore, when all opx exhausted $\phi_2 = 0$, $\phi_f = X/X_e = \chi$. Along this line b, $\chi_f = \chi_e = 1$ is a constant for the fact that all the melting are eutectic melting. Hence along line b, we have

$$\mathcal{C} = \chi \tag{7.10}$$

Now let's turn to another line c separating region IV (super-eutectic melting) and region VI (super-liquidius region). On this line c, all solid has molten $\phi_f = 1$. The resulting $\mathcal{C} = \chi_f = \chi$. Hence the final $\mathcal{H} - \mathcal{C}$ diagram is:



7.2.2. Region II. Two-phase sub-solidus: olivine + opx. The conditions to be in region II are:

$$\mathcal{H} < 0 \tag{7.11}$$

In this phase region, temperature hasn't reached eutectic melting temperature.

$$\phi_2 = \mathcal{C} \quad \phi_1 = 1 - \phi_2 \quad \phi_f = 0 \quad \text{and} \quad \mathcal{T} = \frac{\mathcal{H}}{\phi_1 + \phi_2} = \mathcal{H} < 0 \tag{7.12}$$

7.2.3. Region III. In this three-phase eutectic region, all three phases are present. In this phase region, temperature is exactly at the eutectic temperature,

$$\phi_f = \frac{\mathcal{H}}{L} \quad \phi_2 = \mathcal{C} - \phi_f \chi_f(\mathcal{T}, \mathcal{P}) \quad \phi_1 = 1 - \phi_f - \phi_2 \quad \text{and} \quad \mathcal{T} = 0 \tag{7.13}$$

Where $\chi_f(\mathcal{T}, \mathcal{P}) = \chi_e(\mathcal{P})$, all melting is eutectic melting. Mass fraction is fixed as eutectic mass fraction.

7.2.4. Region IV. In this super-eutectic two-phase region, opx has been exhausted in eutectic melting. The volumetric fractions can be determined as:

$$\phi_2 = 0 \quad \phi_f = \frac{\mathcal{C}}{\chi_f(\mathcal{T}, \mathcal{P})} \tag{7.14}$$

The corresponding temperature should be determined by solving the following equation.

$$F(\mathcal{T}) = \mathcal{H} - \mathcal{T} - \frac{\mathcal{C}}{\chi_f(\mathcal{T}, \mathcal{P})} L = 0 \tag{7.15}$$

Simple Newton method can be applied to solve this equation if we apply nonlinear $\chi_f - \mathcal{T}$ relationship. If we use simple linear relationship between mass fraction and temperature as $\chi_f = \chi_e(1 - \mathcal{T})$, we can obtain a simple quadratic equation:

$$\begin{aligned} (\mathcal{T}_m - \mathcal{T})\mathcal{H} - (\mathcal{T}_m - \mathcal{T})\mathcal{T} - \mathcal{C}L/\chi_e(\mathcal{P}) = 0 \\ \mathcal{T}^2 - (\mathcal{H} + \mathcal{T}_m)\mathcal{T} + \mathcal{T}_m\mathcal{H} - \mathcal{C}L/\chi_e(\mathcal{P}) = 0 \end{aligned} \tag{7.16}$$

7.2.5. Region VI. In this single phase super-liquidus region, we can get volume fractions and temperature easily as:

$$\phi_1 = \phi_2 = 0 \quad \phi_f = 1 \quad \text{and} \quad \mathcal{T} = \mathcal{H} - L \tag{7.17}$$

7.3. Pressure and melting point. Now I take pressure change into account. In general, the higher the pressure, the higher the eutectic melting point. With Clapeyron constant of the value $\gamma = 10^{-7}\mathrm{KPa}^{-1}$ We assume a simple linear relationship between pressure and eutectic melting point.

$$T_e(p) = T_e + \gamma p = T_e + \gamma p$$

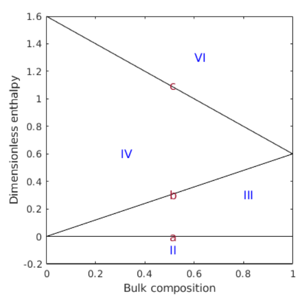
If pressure is replaced with lithostatic pressure, we can relate eutectic melting point with depth.

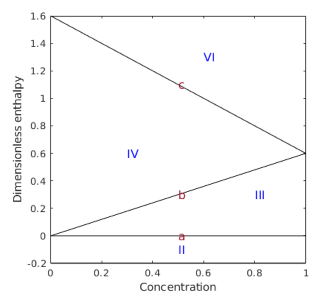
$$T_e(p) = T_e + \gamma p = T_e + \gamma \rho g z$$

An nondimensionlized version can be derived directly by substituting T with $\mathcal{T}(T_1 - T_e) + T_e$.

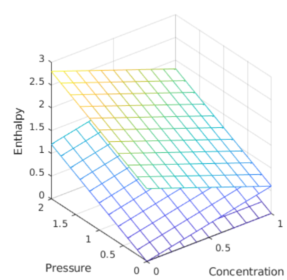
$$\mathcal{T}_e(z) = \frac{\gamma \rho g}{T_1 - T_e} z \tag{7.18}$$

We can plot volumetric fraction and temperature distribution against dimensionless enthalpy and composition as following:





And the phase diagram taking pressure effect into account is displayed as following:



8. Magma migration. For our very first simulation, we would like to simulate the production of magma in a simplified controlled scenerio. I was inspired by the setup described previously in[5]. The simulation will be setup in a chunk of rock ascending with constant upwelling velocity. A linear adiabatic temperature profile is applied here. A linear intial rock composition is also applied here.

8.1. Mathematical model. We are going to show our coupled darcy-stokes semi-degenerate model can solve this problem well. In[5], matrix creeping (Stokes equation) did not directly involved in the simulation due to the fact that boundary condition restricts a constant background upwelling ascending velocity. The solid matrix pressure is therefore strictly lithostatic pressure relating only to depth. In[5], only energy transports while concentration does not participate in transport phenomenon.

$$
\begin{aligned}
\check{\mathbf{v}}_r + \phi_f^{1+\Theta}\nabla(\phi_f^{-1/2}\check{q}_f) &= 0 \\
\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0 \\
\nabla\check{q} - \nabla\cdot\hat{\sigma}(\check{\mathbf{v}}_s) &= -(1-\phi_f)\hat{\mathbf{g}} \\
\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0
\end{aligned}
\tag{8.1}
$$

For simplification, we won't include diffusion into our transport model. We drop the diffusion term here for the fact that diffusion coefficient is relatively small compared to convective phenomenon. On the other hand, eutectic melting has a constant concentration such that gradient of concentration is of zero value. Hence the transport of component 2 (opx) can be expressed in terms of mixture theory as:

$$
\begin{aligned}
\bar{C}_t + \nabla\cdot(\overline{C\mathbf{v}}) &= 0 \\
\bar{C}_t + \nabla\cdot(\mathbf{v}_e^*\bar{C}) &= 0
\end{aligned}
\tag{8.2}
$$

where $\mathbf{v}_e^*$ denotes effective veclocity $\mathbf{v}_e^* = \kappa\mathbf{v}_f + (1-\kappa)\mathbf{v}_s$ and $\kappa = \frac{\phi_f C_f}{\bar{C}}$ where C_f and $\bar{C}$ using values from previous time step. Enthalpy is transported in a similar manner.

$$
\bar{\mathcal{H}}_t + \nabla\cdot(\overline{\mathcal{H}\mathbf{v}} - \bar{K}\nabla\mathcal{T}) = 0 \tag{8.3}
$$

8.1.1. Boundary conditions. In order to solve the proposed darcy-stokes coupled equation, we need appropriate boundary condtions defined around the computational domain for stokes and darcy subproblems. The behavior of solid matrix is governed by Stokes equation. Bernardi-raugel style element is used for this Stokes system, hence we need to provide proper boundary conditions for two dofs per nodal and one additional supplemental flux bubble function for each edge dof. The corresponding boundary condition are shown as follows.

- Prescribe essential boundary condition of constant upwelling ascending velocity at supplying depth $z = z_0$ with $\mathbf{v}_s = \mathbf{W}$.
- Prescribe essential boundary condition to normal velocity and supplemental flux with the value of zero on two sides of the physical domain.
- Prescribe zero traction (free dof) boundary condition (natural boundary condition) to tangantial velocity on two sides of the physical domain.
- Prescribe natural boundary conditoint to normal velocity and supplemental flux on the top side of the physical domain representing a free outlet scenerio.

- Prescribe essential boundary condition to tangential velocity with value zero on the top side of the physical domain.

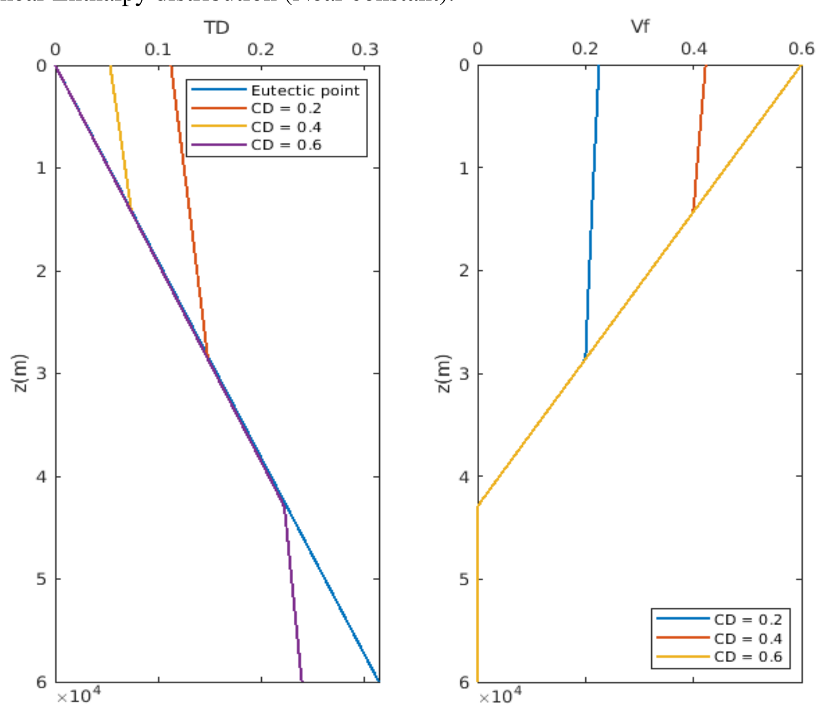
The behavior do molten matrix is governed by Darcy equation. We used BDM sytle element for this Darcy system, hence we just need to provide proper boundary condition for each edges. The corresponding boundary condition are shown as follows:

- Prescribe zero values on two sides of the physical domain.
- Prescribe zero values on the bottom of the physical domain representing no melting enters the computational domain form bottom.
- Prescribe free (natural) boundary condition on the top of the physical domain representing a free outlet of the melt.

8.1.2. Initial conditions. At the same time, we need an initial conditions for our transport equations. In[5], temperature and composition are distributed linearly along against the depth. Unlike [5], we implement eutectic melting model into our simulation. In our situation, temperature and total composition won't be deterministic. Here I make some modification on the setup. Initially, the entire domain was burried under the partial melting region, where temperature was under the sub-solidus region. Melting starts when the entire chunck ascending adiabatically. As pressure continuously lowering, the eutectic point is lowering at the same time. Hence, melting flux supplies from top.

In eutectic melting situation, we cannot impose simple linear temperature directly due to the existence of the eutectic point. Therefore, an appropriate initial condition should be assigned with careful selection.

Linear Enthalpy distribution (Near constant).



8.1.3. Melting depth. As the magma chunk ascending adiabatically, we can define a depth z_m where melting starts. From the plots above, we can conclude that in our model

- Change of dimensionless composition can only affect the separaration of eutection and super-eutection region.

- Change of dimensionless composition will not affect where magam leave eutection melting region.

A fixed melting depth can therefore be calculated despite dimensionless composition distribution. The melting happens when current temperature reaches the eutectic melting temperature. In the situtation of linear dimensionless enthalpy distribution $f(HD) = HD_0 + \eta z$. Assuming lithostatic pressure distribution, the eutectic point at given depth z is: $\mathcal{T}_e = 0 + \gamma * \rho_s g z$. The melting depth is therefore calculated as: $z = \frac{HD_0}{\rho_s g - \eta}$. In this case, the smaller initial dimensionless enthalpy HD_0 the shorter melting depth.

8.2. Interpolation of stokes pressure. We are using piece-wise constant L2 approximation space for stokes and darcy pressure. In order to evaluate point-wise pressure. We are going to present an interpolation using generated piece-wise constant pressure.

8.3. Approximation of physical values on the cell edges. In the process of simulation, plenty of physical values are defined on the gauss points of cell edges. Take κ for example, where $\kappa = \frac{\phi_f c_f}{\bar{c}}$, two phase parameters ϕ_f and c_f are computed with phase package using reconstructed values on both side of the cell edges. The value of $\bar{c}$ are directly obtained by WENO reconstruction. Therefore, two different values κ_{In} and κ_{Out} will be obtained during this computation process, which will be used in the calculation of the numerical flux.

8.4. Nondimensionalization and reterive of original physical variables. We converting scaled and nondimensionalized computational variables back to original physical variables. Note, the characteristic length defined similarly to the definition of compact length.

$$\tilde{\mathbf{v}}_r = \phi^{-1-\Phi}\mathbf{u} \quad , \quad \tilde{q}_f = \phi^{1/2} q_f \quad \text{and} \quad q = q_s + \phi(q_f - q_s)$$

while the pressure potentials are defined as

$$q_f = p_f - \rho_f g z \quad \text{and} \quad q_s = p_s - \rho_f g z$$

and we define reference physical variables as:

$$l_0 = (\frac{k_0 \mu_s}{\mu_f})^{1/2} \quad \text{and} \quad q_0 = \rho_r g l_0 \quad \text{and} \quad w = \frac{k_0 \rho_r g}{\mu_f}$$

Hence we can reterive the two original physical pressure as:

$$p_f = q_0 \hat{q}_f + \rho_f g z$$
$$p_s = \frac{\hat{q} - \phi \hat{q}_f}{1-\phi} q_0 + \rho_f g z$$

where $\hat{q}$ and $\hat{q}_f$ are dimensionless variables computed in the linear system.

Scale	Definition	Typical value	Units
V_0		3.2	cm y^{-1}
l_0	$(\frac{k_0 \mu_s}{\mu_f})^{1/2}$		

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